Cambridge Instruments & Engg. Co.

EST. 1963 · HOWRAH, INDIA

TECHNICAL CATALOGUE · EARTH TESTER · METAL BODY

# CIE/222M

3-terminal and 4-terminal hand-driven earth tester

## Earth resistance and soil resistivity, in the field

Direct measurement of the resistance of earth in power circuits, telecommunications, railway electrification, domestic and industrial electrical installations, lightning protection installations and various other electrical systems. The tester performs equally well in terrain where the conductivity of the soil is low, for example in mountainous areas, and in places where powerful stray earth currents exist. The four-terminal instrument is specially designed for straightforward, approximate soil investigation of the kind needed in agricultural and geological research and in the planning of large civil engineering projects such as tunnels, drainage, sewerage schemes and tube railways.




CIE/222M · ILLUSTRATIVE IMAGE

## 01 HIGHLIGHTS

- ✓ Silent generator with collapsible handle
- ✓ Chuck nut locking arrangement prevents anti-clockwise rotation
- ✓ Anti-magnetic aluminium body with protected metallic flap over the scale window
- ✓ Accuracy conforms to IS 9223:1979
- ✓ Available in 3-terminal and 4-terminal types
- ✓ Centrifugal clutch (governor) for constant output voltage
- ✓ Wide anti-parallax mirror scale
- ✓ Accurate, reliable and quick response
- ✓ Long life and strength against vibration and impact
- ✓ Single range, or multi-range with range selector switch

## 02 TYPICAL APPLICATIONS

- Earth resistance measurement for power circuits and lightning protection systems
- Telecommunications and railway electrification earthing checks
- Domestic and industrial electrical installation earthing
- 4-terminal type: soil resistivity surveys for civil engineering projects (tunnels, drainage, sewerage)



All specifications are subject to change without prior notice. © Cambridge Instruments & Engg. Co. - CIE/222M Technical Catalogue.

## 03 PRINCIPLE AND CONSTRUCTION

### PRINCIPLE

Test current is generated by a smooth and noiseless hand-driven A.C. generator. A high-grade centrifugal governor holds the voltage constant for accurate measurement, so variation in crank speed does not affect the reading.

### CONSTRUCTION

Housed in a sturdy, elegant case of anti-magnetic cast aluminium with a metallic carrying handle. The instrument is dust-free: rubber shoes hold the surface and give it stability, and a protected metallic flap covers the glass window of the scale. The collapsible generator handle folds back flush into the casing when the tester is not in use.

## 04 SALIENT FEATURES

**Immune to stray currents**

Measuring accuracy is not affected by stray alternating current or by the back e.m.f. of earth electrolysis.

**Centrifugal governor**

Holds the voltage constant, so variation in crank speed does not affect the reading.

**3-terminal or 4-terminal**

Three terminals for earth resistance. Four terminals for earth resistance and specific soil resistivity.

**Other resistance measurement**

The instrument can also be used for measuring non-inductive or non-capacitive resistance.

## 05 PERFORMANCE AND TEST SPECIFICATION

| | |
|---|---|
| Standard | IS 9223:1979 |
| Types | 3-terminal (earth resistance); 4-terminal (earth resistance and specific soil resistivity) |
| Generator speed | 160 r.p.m. |
| Ranges | Single range from 0–1 Ω to 0–10,000 Ω (any one range); multi-range with range selector switch; any other range on specific requirement |
| Scale length | 80 mm (approx.) |
| Accuracy | ±5% (as per IS 9223:1979) |
| High voltage test | 2 kV A.C. (r.m.s.) for 1 minute between the electrical circuit and the containing case |
| Insulation resistance of the instrument | Better than 25 MΩ at 500 V D.C. between the electrical circuit and the containing case |



All specifications are subject to change without prior notice. © Cambridge Instruments & Engg. Co. - CIE/222M Technical Catalogue.

## 06 CONSTRUCTION AND PHYSICAL SPECIFICATION

| | |
|---|---|
| Generator | Hand-driven A.C. generator with centrifugal governor |
| Body | Anti-magnetic cast aluminium case with metallic carrying handle and protected metallic flap over the scale window |
| Weight | 2.5 kg (approx.) |
| Dimensions | 110 × 128 × 190 mm (approx.) |

## 07 SINGLE RANGE OPTIONS

| RANGE BAND | SELECTION |
|---|---|
| 0–1 Ω to 0–5 Ω | Any range within this band |
| 0–10 Ω to 0–100 Ω | Any range within this band |
| 0–200 Ω to 0–10,000 Ω | Any range within this band |

*A single-range instrument has **one** range: choose any one range from 0–1 Ω up to 0–10,000 Ω within the three bands above, in either the 3-terminal or the 4-terminal type.*

## 08 DUAL RANGE OPTIONS

| CONFIGURATION | FULL-SCALE RANGES (Ω) |
|---|---|
| Dual range | 0–1 / 10 · 0–2 / 10 · 0–3 / 30 · 0–10 / 100 |

## 09 TRIPLE, FOUR AND FIVE RANGE OPTIONS

| CONFIGURATION | FULL-SCALE RANGES (Ω) |
|---|---|
| Triple range | 0–1 / 10 / 100 · 0–2 / 10 / 100 · 0–2 / 20 / 200 · 0–3 / 30 / 300 · 0–10 / 100 / 1000 |
| Four range | 0–1 / 10 / 100 / 1000 · 0–2 / 10 / 100 / 1000 · 0–10 / 100 / 1000 / 10,000 |
| Five range | 0–1 / 10 / 100 / 1000 / 10,000 · 0–2 / 10 / 100 / 1000 / 10,000 |

*Multi-range instruments are provided with a range selector switch, to be chosen from 0–1 Ω to 10,000 Ω. Any other range can be manufactured against specific requirements.*



All specifications are subject to change without prior notice. © Cambridge Instruments & Engg. Co. - CIE/222M Technical Catalogue.

## 10 3-TERMINAL AND 4-TERMINAL TYPES

| | 3-TERMINAL TYPE | 4-TERMINAL TYPE |
|---|---|---|
| **Measures** | Earth resistance | Earth resistance and specific soil resistivity |
| **Typical use** | Earth resistance of power circuits, telecommunications, railway electrification, installations and lightning protection systems | As the 3-terminal type, plus soil investigation for agricultural and geological research and civil engineering projects |
| **Test kit (optional)** | 2 M.S. spikes (45 cm length), 3 rolls of flexible wire on spool (10 m, 20 m, 30 m), 1 hammer, all in one carrying bag | 4 M.S. spikes (45 cm length), 4 rolls of flexible wire on spool (10 m, 20 m, 20 m, 30 m), 1 hammer, all in one carrying bag |
| **Ranges** | All ranges listed in this catalogue | All ranges listed in this catalogue |

## 11 ACCESSORY OPTIONS (AT EXTRA COST)

- Testing kit for the 3-terminal or the 4-terminal type (contents in the table above)
- Carrying case for earth tester

## 12 HOW TO ORDER

To order, quote the **model**, the **terminal type** and the **range**, for example `CIE/222M, 4-terminal, 0–1 / 10 / 100 Ω`. Any range not listed can be manufactured against specific requirements.



All specifications are subject to change without prior notice. © Cambridge Instruments & Engg. Co. - CIE/222M Technical Catalogue.